

Lab - 5 : Modelling data with power laws (Pareto's law and Zipf's law)

Krutarth Kadia (202201475)* and Het Shah (202201515)[†]
*Dhirubhai Ambani Institute of Information & Communication Technology,
 Gandhinagar, Gujarat 382007, India
 CS302, Modelling and Simulation*

I. PARETO DISTRIBUTION OF WEALTH IN INDIA

A. Model

To model the power law that the data follow, apply the function

$$N(x) = A + Bx^{-\alpha} \quad (1)$$

Given that x is the amount of wealth and $N(x)$ is the frequency distribution of wealth holders.

B. Results

Fig. 1 shows pareto distribution of wealth in India.

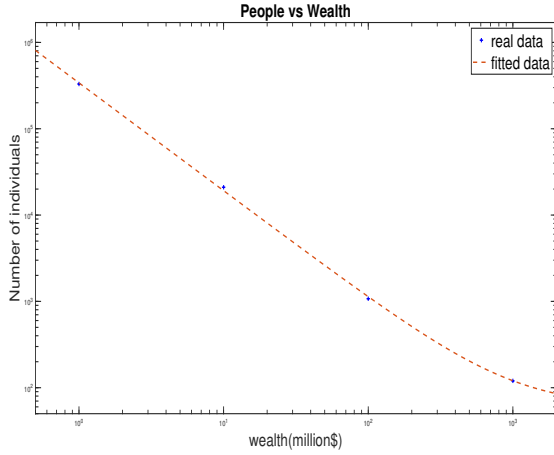


FIG. 1: Here $A=60$, $B = 340000$ and $\alpha = 5/4$.

II. ZIPF'S LAW IN THE DEPENDENCY NETWORK OF DEBIAN

A. Model

The global power law distribution is given by,

$$\phi(x) = [\eta + (\frac{x + \lambda}{c})^{-\mu\alpha}]^{-1/\mu} \quad (2)$$

in which α is a power-law exponent, μ is a nonlinear saturation exponent, η is a “tuning” parameter for non-linearity, and λ is another parameter that is instrumental in setting a limiting scale for the poorly connected nodes.

With $\mu = -1$ (implying a power-law in the distribution) and with $\alpha = -2$ (implying that the power-law is specifically Zipf's law), the saturation properties of the network (for any value of λ and η) can be abstracted from equation (2) as

$$\phi(x) = \eta + (\frac{c}{x + \lambda})^2 \quad (3)$$

B. Results

Fig. 2 shows the network of incoming links in the Etch release.

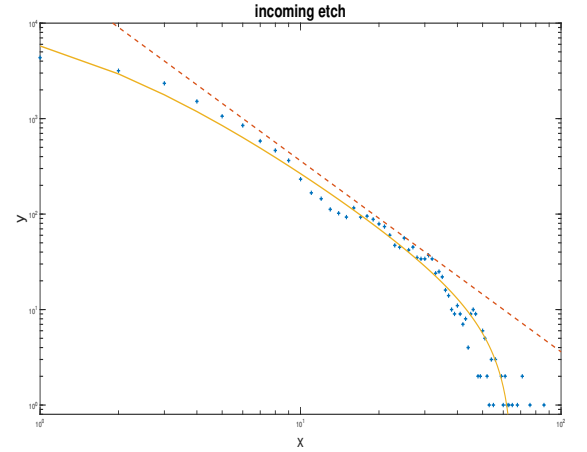


FIG. 2: Here $\alpha = -2$, $\mu = -1$, $\eta = -8$, $\lambda = 1.5$ and $c = 190$.

*Electronic address: 202201475@daiict.ac.in

[†]Electronic address: 202201515@daiict.ac.in

Fig. 3 shows the network of outgoing links in the Etch release.

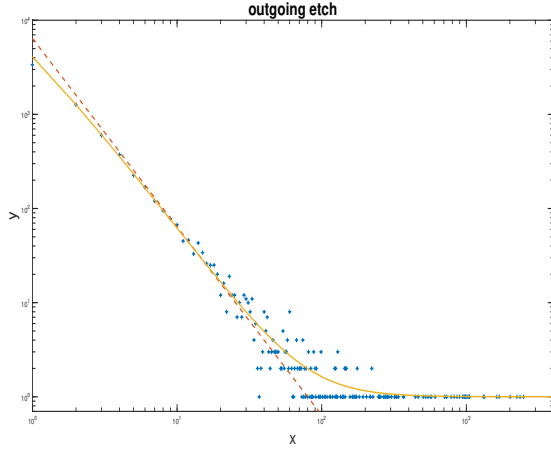


FIG. 3: Here $\alpha = -2$, $\mu = -1$, $\eta = 1$, $\lambda = 0.25$ and the data is fitted for $c = 80$. A solitary top node is to be seen for $x = 9025$.

Fig. 5 shows the network of outgoing links in the Lenny release.

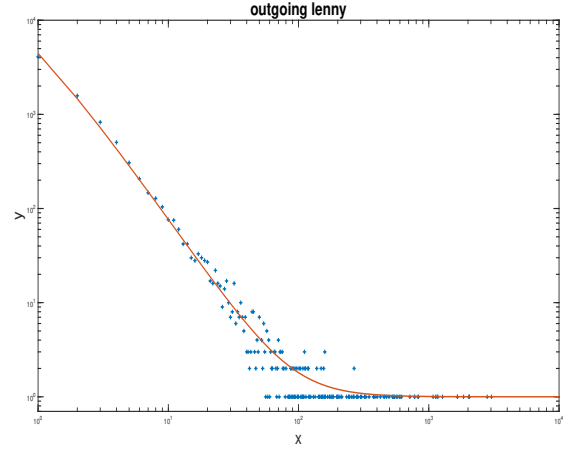


FIG. 5: Here $\alpha = -2$, $\mu = -1$, $\eta = 1$, $\lambda = 0.35$ and the data is fitted for $c = 90$. A solitary top node is to be seen for $x = 10446$.

Fig. 4 shows the network of incoming links in the Lenny release.

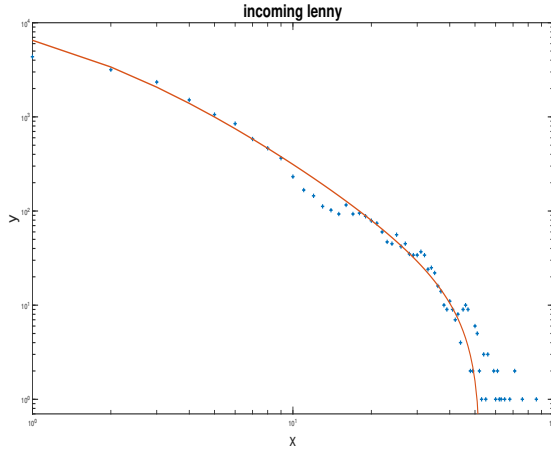


FIG. 4: Here $\alpha = -2$, $\mu = -1$, $\eta = -15$, $\lambda = 1.6$ and $c = 210$.

Fig. 6 shows the network of incoming links in the Squeeze release.

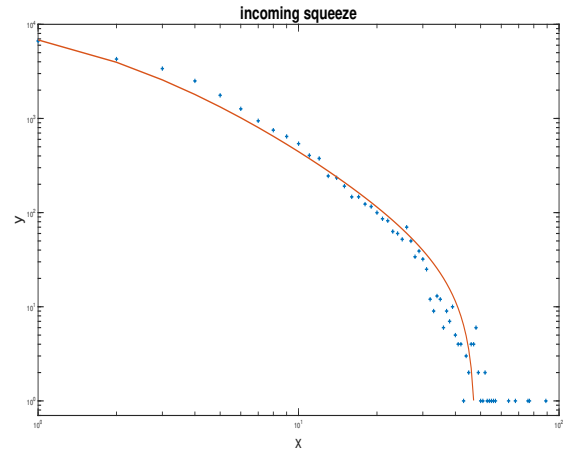


FIG. 6: Here $\eta = -28$, $\lambda = 2.2$ and $c = 265$.

Fig. 7 shows the network of outgoing links in the Squeeze release.

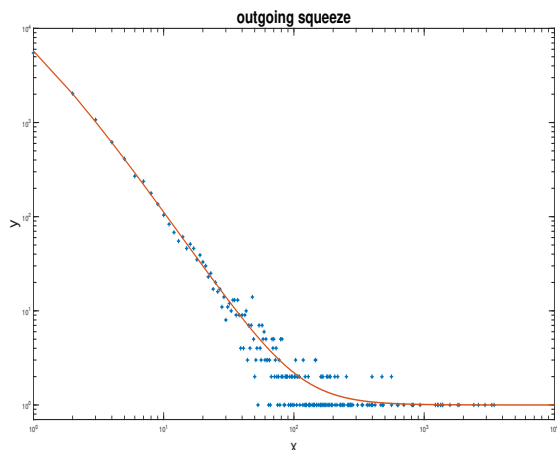


FIG. 7: Here $\eta = 1$, $\lambda = 0.45$ and the data is fitted for $c = 110$. The richest node in this distribution has 12470 links.

III. CONCLUSIONS

- The Pareto distribution of wealth in India, based on 2021 data, highlights a pronounced wealth gap, where a minority possesses the majority of wealth.

This pattern, evident in our graph, aligns with the characteristics of a power law distribution.

- Small number of packages have a very high frequency of dependencies, while the vast majority of packages have relatively few dependencies. Packages that are very important and widely used in software tend to have a lot of other packages depending on them directly, and they are often more likely to have strong relationships with those packages. This is because important packages are relied on by many other packages, leading to a higher number of connections to them.
- The frequency of packages in the Debian dependency network follows a power law distribution, which is a mathematical pattern that is closely related to Zipf's law.

[1] Rajiv Nair, G. Nagarjuna, and Arnab K. Ray, '*Finite-Size Effects in the Dependency Networks of Free and*

Open-Source Software'.